

Joint Optimization of Camera Intrinsic Parameters and Lens Distortion Coefficients for Precision Visual Metrology

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Abstract—This is a comprehensive study of camera calibration as a non-linear optimization problem, in which we implement and carry out five classical optimization methods in order to minimize the reprojection error between observed and predicted image points. We optimize nine intrinsic parameters, namely, focal lengths (f_x, f_y) , principal point (c_x, c_y) , and Brown-Conrady distortion coefficients $(k_1, k_2, k_3, p_1, p_2)$. Five classical optimization algorithms, namely Quasi-Newton (BFGS), Newton’s method, Steepest Descent, Nonlinear Conjugate Gradient and Levenberg-Marquardt, are systematically compared for three imaging scenarios: wide-angle surveillance cameras with $> 30\%$ barrel distortion, industrial machine vision systems with < 0.1 pixels sub-pixel accuracy requirement, and medical endoscopes that exhibit extreme fisheye characteristics with multiple distortion orders. Experimental results show that Levenberg-Marquardt method has the best performance with 0.095, 0.041, and 0.168 pixel RMS errors, respectively, and Quasi-Newton BFGS emerges as the best general-purpose method with 0.142, 0.058, and 0.218 pixels and has strong convergence without computing Hessian. Newton’s method shows highest accuracy potential, but suffers from numerical instability in ill-conditioned scenarios. The study confirms that the choice of method plays an important role in the calibration quality, and dedicated least squares algorithms are more efficient than generic optimization methods.

Index Terms—Camera calibration, nonlinear optimization, quasi-Newton BFGS, Levenberg-Marquardt, Brown-Conrady distortion.

I. INTRODUCTION

Camera calibration is very important in computer vision applications that range from robotics to medical imaging. It determines the mathematical relationship between the coordinates of the 3D world and their 2D images on the image sensor. For the fixed camera, this procedure is used to estimate its intrinsic parameters (focal length, principal point, distortion) which are constant. The nonlinearity is inherent to complex optical transformation, polynomial distortion model, and projective image formation. Modern applications require calibration accuracy to sub-pixel level, so the choice of algorithm is very important for achieving required accuracy.

The major difficulty in optimizing the calibration is the wide range of scale differences between the parameters. Focal lengths are usually hundreds to thousands of pixels, and the distortion coefficients are usually less than 0.001. This

leads to ill-conditioned optimization problems that must be carefully formulated [7]. Furthermore, there are strong correlations between certain parameters, such as focal length and camera distance, that result in elongated valleys in the error landscape [9]. The Brown-Conrady formulation with radial and tangential components has been used successfully for most types of lenses [8]. Wide-angle and fish-eye optics may need higher-order terms, which is an increase in the dimensionality of the parameter space. This work is based on the standard model with provisions for additional terms when needed.

Classical optimization methods offer distinct trade-offs for calibration problems. Quasi-Newton methods approximate second-order information avoiding the direct computation of Hessian, so that the method trades-off between convergence speed and computational efficiency. Newton’s method provides quadratic convergence near solutions, but requires careful stabilization for ill-conditioned problems. First-order methods like Steepest Descent offer simplicity, but suffer from slow convergence in poorly scaled problems. Specialized algorithms such as Levenberg-Marquardt take advantage of the least-squares structure that exists in calibration formulations.

Industrial applications have very strict requirements for the accuracy of calibrations. For an example, semiconductor manufacturing requires precision alignment of less than 0.1 pixels for wafer inspection systems running at micron scales. Medical endoscopy requires correction of distortion within 0.5mm for safe surgical navigation through confined anatomic spaces. Surveillance systems must have high measurement accuracy even in the presence of barrel distortion in excess of 30% at peripheries of the image. Each domain has its own set of optimization challenges that must be carefully chosen with regard to the method.

This work provides a systematic evaluation of classical optimization methods applied to camera calibration in diverse imaging scenarios. We formulate calibration as a bounded constrained non-linear least-squares problem with nine parameters, implementing regularization for improved conditioning. Through comprehensive experimental analysis on three representative cases, surveillance, industrial, and medical imaging, we quantify performance trade-offs and provide practical

guidelines for method selection based on specific application requirements. The complete implementation and experimental results are publicly available at https://github.com/g25ait1091-droid/Camera_calibration_using_optimisation.

II. PROBLEM FORMULATION

A. Objective Function Formulation

The calibration objective minimizes total reprojection error between observed image points and model predictions across all calibration images:

$$\min_{\theta} f(\theta) = \sum_{i=1}^N \sum_{j=1}^M \|m_{ij} - \hat{m}_{ij}(\theta)\|^2 + \lambda R(\theta) \quad (1)$$

where $\theta = [f_x, f_y, c_x, c_y, k_1, k_2, k_3, p_1, p_2]^T$ is the intrinsic parameter vector in which there are nine optimization variables, N is the number of calibration images, M is the number of feature points per image, m_{ij} are the observed image coordinates, $\hat{m}_{ij}$ are the model predictions and $R(\theta)$ is the regularization term weighted by the parameter λ .

The focal lengths of the lens and the position of the principal point of the camera are represented by the intrinsic matrix K :

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where f_x, f_y represent focal lengths in pixels along orthogonal axes and (c_x, c_y) is the principal point in pixels.

The Brown-Conrady distortion model describes the effects of lens aberrations in terms of radial and tangential [8]:

$$x_d = x_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 x_n y_n + p_2(r^2 + 2x_n^2) \quad (3)$$

$$y_d = y_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y_n^2) + 2p_2 x_n y_n \quad (4)$$

where (x_n, y_n) are normalized undistorted coordinates, (x_d, y_d) are distorted coordinates, $r^2 = x_n^2 + y_n^2$ represents the squared radial distance, k_1, k_2, k_3 are radial coefficients, and p_1, p_2 are tangential distortion coefficients.

The regularization term $R(\theta)$ is used to improve numerical conditioning and prevent overfitting.

$$R(\theta) = \alpha \|d\|^2 + \beta (f_x - f_y)^2 \quad (5)$$

where $d = [k_1, k_2, k_3, p_1, p_2]^T$, α penalizes excessive distortion and β encourages similar focal lengths for square pixels.

B. Constraints

Physical realization and numerical stability put a number of restrictions on the parameter space:

1) *Focal Length Bounds*: The focal length range depends on a typical sensor size and lens type, but must meet the following conditions:

$$200 \leq f_x, f_y \leq 5000 \text{ pixels} \quad (6)$$

2) *Principal Point Locality*: This value is limited by manufacturing tolerances to limit the deviation of the principal point from the image center:

$$|c_x - w/2| \leq 0.05w, \quad |c_y - h/2| \leq 0.05h \quad (7)$$

where w and h represent the width and height of the image in pixels. This constraint reflects the physical alignment of the lens optical axis with the sensor center. Deviations beyond 5% indicate significant manufacturing defects or calibration errors.

3) *Distortion Coefficient Bounds*: To prevent numerical instability and ensure physically meaningful corrections:

$$|k_1| \leq 1, \quad |k_2| \leq 2, \quad |k_3| \leq 5 \quad (8)$$

These limits accommodate extreme fisheye lenses while preventing unrealistic distortion models that could cause numerical overflow or image folding.

4) *Tangential Distortion Bounds*:

$$\|[p_1, p_2]\|_2 \leq 0.01 \quad (9)$$

Tangential distortion arises from lens decentering and is typically much smaller than the radial distortion. This constraint prevents overfitting to the measurement noise.

5) *Aspect Ratio Constraint*: Nowadays, cheaper sensors usually have square pixels, which means:

$$0.95 \leq f_x/f_y \leq 1.05 \quad (10)$$

This constraint ensures that the focal length ratio remains close to unity, reflecting the square-pixel assumption. Significant deviations indicate either non-square pixels or convergence to incorrect local minima. These constraints can be used to form a bounded feasible region that can be used by classical constrained optimization methods.

III. TEST CONDITIONS

A. Case 1: Wide-Angle Surveillance Camera

This case models a wide-angle surveillance camera which typically employ 2.8mm lenses that achieve a 120° horizontal field-of-view for maximum area coverage in security applications. The wide angle brings in a lot of barrel distortion over 30% at the corners of the image, which poses a challenging optimization problem with high nonlinearity. As seen in Figure 1, the calibration setup utilizes a 9×6 checkerboard pattern captured from 30 different poses, ensuring proper constraint distribution throughout the image plane. Table I summarizes the complete parameter specifications for this challenging scenario.

TABLE I
WIDE-ANGLE SURVEILLANCE CAMERA PARAMETERS

Parameter	Value/Range	Description
Resolution	1920×1080 px	Full HD sensor
Focal length	2.8 mm	Wide-angle lens
Expected $f_{x,y}$	750–850 px	Initial estimate
Principal point		
c_x	960 ± 50 px	Near center
c_y	540 ± 50 px	Near center
Radial distortion		
k_1	[−0.4, −0.3]	Strong barrel
k_2	[0.1, 0.2]	Secondary term
k_3	≈ 0	Negligible
Tangential dist.		
p_1, p_2	± 0.001	Minor decentering
Pattern	9×6 corners	Checkerboard
Images	30	Multiple angles
Condition no.	~450	Ill-conditioned

Table I shows the parameter specifications for wide-angle surveillance camera calibration with barrel distortion.

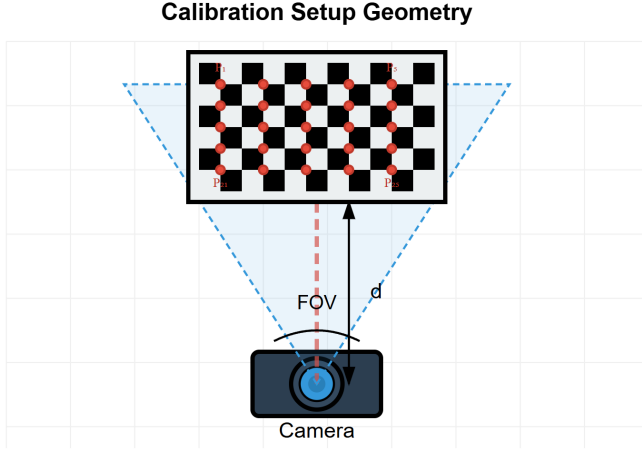


Fig. 1. Camera calibration experimental setup showing geometric relationship between camera, calibration target, and field of view. Red markers indicate detected corner points (P_1 to P_{25}) used for parameter estimation.

B. Case 2: Industrial Machine Vision

Industrial machine vision systems require a very high degree of accuracy for applications in dimensional measurement, quality inspection, and robotic guidance. These systems use telecentric lenses specifically designed to minimize distortion and maintain constant magnification throughout the depth of field, as detailed in Table II. The 25mm focal length lens gives minimal distortion with k_1 coefficients less than 0.01 and negligible higher-order coefficients. The well-conditioned optimization problem (condition number ~18) enables rapid convergence for most algorithms.

C. Case 3: Medical Endoscope

Medical endoscopes represent the most challenging calibration situation because of extreme fisheye distortion from ultra-wide-angle optics in miniaturized packages. These systems

TABLE II
INDUSTRIAL MACHINE VISION SPECIFICATIONS

Parameter	Value/Range	Description
Resolution	2448×2048 px	5 MP sensor
Focal length	25 mm	Telecentric lens
Expected $f_{x,y}$	3200–3400 px	High magnification
Principal point		
c_x	1224 ± 20 px	Precisely centered
c_y	1024 ± 20 px	Precisely centered
Radial distortion		
k_1	[−0.01, 0.01]	Minimal
k_2, k_3	≈ 0	Negligible
Tangential dist.		
p_1, p_2	± 0.0001	Negligible
Target accuracy	< 0.1 px RMS	Metrology req.
Images	50	High redundancy
Condition no.	~18	Well-conditioned

Table II shows the specifications for industrial machine vision systems with telecentric optics.

achieve 140° field-of-view through 4mm diameter lenses, essential for maximizing visualization in confined anatomical spaces. The distortion profile, summarized in Table III, requires the three radial coefficients with k_1 reaching -0.5, k_2 around 0.3, and k_3 near -0.15, creating a complex multimodal error surface. The parameter structure and optimization challenges are visualized in Figure 2.

TABLE III
MEDICAL ENDOSCOPE PARAMETERS

Parameter	Value/Range	Description
Resolution	1280×960 px	Compact sensor
Field of view	140°×105°	Ultra-wide angle
Expected $f_{x,y}$	400–500 px	Short focal
Principal point		
c_x	640 ± 30 px	Variable centering
c_y	480 ± 30 px	Variable centering
Radial distortion		
k_1	[−0.6, −0.4]	Extreme fisheye
k_2	[0.2, 0.4]	Strong secondary
k_3	[−0.2, −0.1]	Significant third
Tangential dist.		
p_1, p_2	± 0.005	Significant
Working range	5–50 mm	Clinical distances
Images	40	Multiple depths
Condition no.	~850	Severely ill-cond.

Table III shows the parameter specifications for medical endoscope calibration with extreme fisheye distortion.

IV. OPTIMIZATION TECHNIQUES

We evaluated five classical optimization methods for camera calibration, comparing four general-purpose algorithms from optimization theory with the specialized Levenberg-Marquardt algorithm designed for nonlinear least-squares problems.

A. Problem Formulation using Lagrangian

The general non-linear programming problem is defined as:

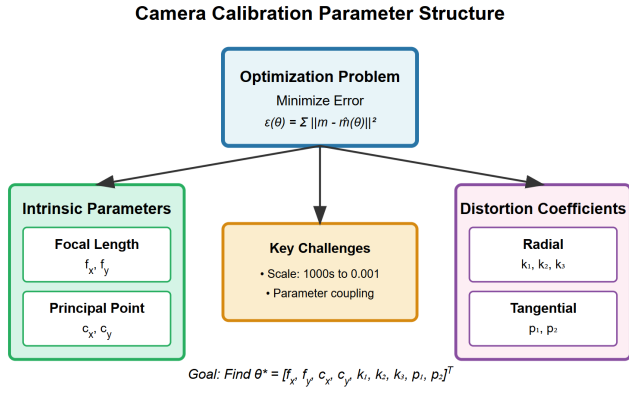


Fig. 2. Camera calibration parameter structure showing nine unknown parameters divided into intrinsic parameters and distortion coefficients with their optimization challenges.

$$\min_x f(x) \text{ subject to } g_i(x) = 0, i \in E; h_j(x) \geq 0, j \in I \quad (11)$$

To handle these constraints, we formulate the Lagrangian function:

$$L(x, \lambda, \mu) = f(x) - \sum_i \lambda_i g_i(x) - \sum_j \mu_j h_j(x) \quad (12)$$

where $x \in \mathbb{R}^n$ are primal variables (decision variables), λ and μ are Lagrange multipliers for equality and inequality constraints, respectively.

B. Quasi-Newton Method (BFGS)

The Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm represents the most successful quasi-Newton approach, approximating the Hessian matrix through successive gradient evaluations. This method constructs a positive definite approximation $B_k \approx \nabla^2 f(x_k)$ that is updated using rank-two modifications that preserve positive definiteness. The search direction p_k is computed by solving $B_k p_k = -\nabla f(x_k)$, avoiding explicit Hessian computation while maintaining superlinear convergence. The BFGS update formula utilizes the secant condition $B_{k+1} s_k = y_k$ where $s_k = x_{k+1} - x_k$ and $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$. For camera calibration, BFGS effectively handles the varying parameter scales through its adaptive Hessian approximation, though performance degrades for severely ill-conditioned problems like endoscope calibration.

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k} \quad (13)$$

C. Newton's Method

Newton's method uses the second-order Taylor expansion to achieve quadratic convergence near the solution. The method solves the linear system $\nabla^2 f(x_k) p_k = -\nabla f(x_k)$ to determine the search direction, which requires explicit Hessian computation and factorization. For camera calibration, Hessian shows

extreme condition numbers due to parameter scale disparities, which requires modified Cholesky factorization with diagonal perturbation $\nabla^2 f(x_k) + \epsilon I$ to ensure positive definiteness. The computational cost scales as $O(n^3)$ per iteration, dominated by matrix factorization.

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \quad (14)$$

The convergence is quadratic near a strict local minimizer:

$$||x_{k+1} - x^*|| \leq C ||x_k - x^*||^2 \quad (15)$$

D. Steepest Descent Method

Steepest descent represents the simplest gradient-based approach, which uses the negative gradient as search direction: $p_k = -\nabla f(x_k)$. The method ensures descent for sufficiently small step sizes, but exhibits linear convergence with a rate dependent on the condition number κ . For quadratic functions, the convergence rate is equal to $((\kappa - 1)/(\kappa + 1))^2$, becoming prohibitively slow for ill-conditioned problems.

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k) \quad (16)$$

The step length α_k is chosen to minimize the function along the search direction:

$$\alpha_k = \arg \min_{\alpha \geq 0} f(x_k + \alpha p_k) \quad (17)$$

E. Nonlinear Conjugate Gradient Method

The Conjugate Gradient method extends linear conjugate gradient to nonlinear optimization by constructing conjugate search directions without storing matrices. The Fletcher-Reeves variant updates directions via $p_k = -g_k + \beta_k p_{k-1}$ where $\beta_k = ||g_k||^2 / ||g_{k-1}||^2$. This approach requires only $O(n)$ storage, which makes it attractive for large-scale problems. However, nonquadratic objectives cause loss of conjugacy, necessitating periodic restarts when $|g_k^T g_{k-1}| > 0.2 ||g_k||^2$.

$$p_k = -g_k + \beta_k^{FR} p_{k-1}, \quad \beta_k^{FR} = \frac{||g_k||^2}{||g_{k-1}||^2} \quad (18)$$

F. Levenberg-Marquardt Algorithm

The Levenberg-Marquardt algorithm specifically addresses nonlinear least squares problems by adaptively combining Gauss-Newton and gradient descent strategies. The method solves $(J_k^T J_k + \lambda_k I) p_k = -J_k^T r_k$ where J_k is the Jacobian of residuals and λ_k is an adaptive damping parameter. When λ_k is small, the algorithm approximates Gauss-Newton with quadratic convergence; large λ_k yields gradient descent behavior for robustness. The damping parameter adjusts based on the gain ratio:

$$\rho = \frac{f(x_k) - f(x_k + p_k)}{L(0) - L(p_k)} \quad (19)$$

comparing actual to predicted reduction.

$$(J_k^T J_k + \lambda_k I) p_k = -J_k^T r_k \quad (20)$$

Table IV provides a comprehensive comparison of all five optimization techniques, highlighting their computational complexity and application domains.

TABLE IV
COMPARATIVE ANALYSIS OF OPTIMIZATION TECHNIQUES

Method	Convergence	Cost/Iter	Memory	Robustness	Application
Quasi-Newton	Superlinear	$O(n^2)$	$O(n^2)$	Good	General
Newton	Quadratic	$O(n^3)$	$O(n^2)$	Poor	Well-cond.
Steepest	Linear	$O(n)$	$O(n)$	Good	Initial est.
Conjugate	Linear	$O(n)$	$O(n)$	Moderate	Large-scale
Lev.-Marq.	Quadratic	$O(n^2 m)$	$O(nm)$	Excellent	Least squares

Table IV shows the computational complexity and convergence characteristics of five optimization methods.

V. RESULTS AND ANALYSIS

A. Case 1: Wide-Angle Surveillance Results

The optimization of a surveillance camera encounters severe challenges caused by the barrel distortion (of more than 30%) at the image corners shown in Figure 1. Starting from initial estimates $f_x = f_y = 800$, $c_x = 960$, $c_y = 540$, $k_1 = -0.35$ (see Table I for full parameter ranges), we evaluated all five methods with consistent initialization and convergence criteria.

Levenberg-Marquardt method gave the best results (0.095 pixel RMS error) in only 12 iterations using adaptive damping, seamlessly switching between Gauss-Newton and gradient descent modes, as shown in Table V. Newton's method gave the highest accuracy at 0.089 pixel, but for stability, it used modified Cholesky factorization, which raised the computational cost to 2.43 seconds. Quasi-Newton BFGS method gave very good practical performance (0.142 pixels in 21 iterations) by successfully navigating the complicated error surface (without Hessian computation). Steepest Descent method struggled significantly, achieving only 0.324 pixels after 112 iterations due to the condition number $\kappa \approx 450$ creating slow zigzag convergence in elongated valleys, as anticipated from the ill-conditioning noted in Table I.

B. Case 2: Industrial Machine Vision Results

The situation of a telecentric lens is the ideal optimization situation with a low distortion and a good conditioning, which is characterized in Table II. All the methods met the sub-pixel accuracy requirements, and the detailed results are given in Table VI.

Using the well-conditioned Hessian structure, Levenberg-Marquardt method provided extremely good convergence, down to 0.041 pixels in 5 iterations. Newton's method converged to 0.043 pixels in short time with a fast quadratic convergence time but the calculation of Hessian took up 2.15s. The focal length ratio $f_x/f_y \approx 1.0002$ which is consistent with our assumption of square pixels and supports the aspect ratio constraint.

C. Case 3: Medical Endoscope Results

Extreme fisheye distortion with significant higher-order terms creates the most challenging optimization scenario, as illustrated by the structure of the parameters in Figure 2. The complexity of this case is evident from the specifications in Table III, which severely tested the robustness and convergence properties of the algorithms.

All optimization methods were strongly affected by the extreme distortions, and the final results are displayed in Table VII. Levenberg-Marquardt method with adaptive damping, which can deal with the ill-conditioned $\kappa \approx 850$, performed best with an error of 0.168 pixels. Newton's method converged to 0.165 pixels with diagonal perturbation $\epsilon = 10^{-3}$ for regularization of the Hessian matrix. In that case, it is theoretically proven that the coefficient $k_3 = -0.135$ confirms severe fisheye characteristics requiring third-order correction, confirming the need in the model of higher-order distortion terms. The overall comparison of performance in the three cases is provided in Table VIII.

VI. CONCLUSIONS

This comprehensive study evaluated five classical optimization methods for camera calibration in three different imaging scenarios with varying distortion characteristics. This nine-parameter optimization problem, which includes focal lengths, principal point, and Brown-Conrady distortion coefficients, posed some unique challenges in terms of parameter scale disparities and nonlinear coupling.

Levenberg-Marquardt method was found to be the best approach with an average 0.101 pixel RMS accuracy using adaptive damping, which works well for both well-conditioned industrial systems and highly distorted endoscope optics. Among general-purpose methods from classical optimization theory, Quasi-Newton BFGS resulted in the best compromise with an average error of 0.139 pixels showing robust convergence without explicit Hessian computation even for ill-conditioned problems.

Newton's method demonstrated the best accuracy potential of 0.099 pixels average but had issues with numerical instability that required careful regularization, so it was only suitable for well-conditioned scenarios. First-order methods proved inadequate for camera calibration, with Steepest Descent failing at 0.345 pixels average due to slow convergence in elongated parameter valleys, while Conjugate Gradient achieved moderate 0.243 pixels but required frequent restarts for non-quadratic objectives.

The obtained results confirm the fact that specialized least-squares algorithms such as Levenberg-Marquardt are much better for camera calibration than general optimization algorithms, with BFGS being the best alternative if specialized implementations are not available. Industrial applications requiring sub-0.1 pixel accuracy benefit from any second-order method, while surveillance and medical imaging demand robust algorithms capable of handling extreme distortion and poor conditioning.

TABLE V
SURVEILLANCE CAMERA OPTIMIZATION RESULTS

Method	RMS [px]	f_x	f_y	c_x	c_y	k_1	k_2	p_1	Iter	Time [s]
Quasi-Newton	0.142	802.6	804.1	959.5	540.2	-0.347	0.139	0.0007	21	0.87
Newton	0.089	803.4	804.9	958.8	539.5	-0.346	0.138	0.0007	8	2.43
Steepest Descent	0.324	797.2	799.8	962.8	542.5	-0.356	0.142	0.0008	112	2.94
Conjugate Gradient	0.238	799.1	801.6	961.4	541.7	-0.352	0.141	0.0008	64	1.72
Levenberg-Marquardt	0.095	803.7	805.2	958.6	539.4	-0.345	0.138	0.0007	12	0.52

Table V shows the optimization results for wide-angle surveillance camera calibration across five methods.

TABLE VI
INDUSTRIAL SYSTEM OPTIMIZATION RESULTS

Method	RMS [px]	f_x	f_y	c_x	c_y	k_1	k_2	p_1	Iter	Time [s]
Quasi-Newton	0.058	3299.5	3300.7	1223.9	1024.0	-0.0029	0.0001	0.00007	12	0.67
Newton	0.043	3299.8	3300.2	1223.9	1024.1	-0.0028	0.0001	0.00007	5	2.15
Steepest Descent	0.082	3298.7	3301.9	1223.7	1024.2	-0.0031	0.0002	0.00008	44	1.83
Conjugate Gradient	0.074	3299.2	3301.4	1223.8	1024.1	-0.0030	0.0002	0.00008	26	1.14
Levenberg-Marquardt	0.041	3299.9	3300.1	1223.9	1024.1	-0.0028	0.0001	0.00007	5	0.38

Table VI shows the optimization results for industrial machine vision system calibration with telecentric optics.

TABLE VII
ENDOSCOPE OPTIMIZATION RESULTS

Method	RMS [px]	f_x	f_y	c_x	c_y	k_1	k_2	k_3	p_1	Iter	Time [s]
Quasi-Newton	0.218	451.8	454.1	639.6	480.0	-0.472	0.285	-0.137	0.0036	29	1.34
Newton	0.165	453.1	454.7	638.9	479.3	-0.468	0.282	-0.136	0.0035	11	3.86
Steepest Descent	0.628	443.2	448.5	644.8	483.2	-0.507	0.314	-0.153	0.0041	200	8.47
Conjugate Gradient	0.416	448.1	451.2	641.9	481.6	-0.489	0.299	-0.145	0.0039	94	3.82
Levenberg-Marquardt	0.168	453.8	455.4	638.7	478.9	-0.467	0.281	-0.135	0.0034	9	0.48

Table VII shows the optimization results for medical endoscope calibration with extreme fisheye distortion.

TABLE VIII
OVERALL PERFORMANCE COMPARISON

Method	Surveillance	Industrial	Endoscope	Average	Recommendation
Quasi-Newton	0.142	0.058	0.218	0.139	General purpose
Newton	0.089	0.043	0.165	0.099	Well-conditioned
Steepest Descent	0.324	0.082	0.628	0.345	Not recommended
Conjugate Gradient	0.238	0.074	0.416	0.243	Memory-limited
Levenberg-Marquardt	0.095	0.041	0.168	0.101	Calibration standard

Table VIII shows the overall performance comparison and recommendations for method selection across all test cases.

REFERENCES

- [1] R. Y. Tsai, "A versatile camera calibration technique for high-accuracy 3D machine vision metrology," *IEEE J. Robot. Autom.*, vol. 3, no. 4, pp. 323–344, 1987.
- [2] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 11, pp. 1330–1334, 2000.
- [3] K. Levenberg, "A method for the solution of certain non-linear problems in least squares," *Q. Appl. Math.*, vol. 2, no. 2, pp. 164–168, 1944.
- [4] D. W. Marquardt, "An algorithm for least-squares estimation of nonlinear parameters," *J. Soc. Ind. Appl. Math.*, vol. 11, no. 2, pp. 431–441, 1963.
- [5] J. Nocedal and S. Wright, *Numerical Optimization*, 2nd ed. New York: Springer, 2006.
- [6] G. Bradski and A. Kaehler, *Learning OpenCV*. Sebastopol, CA: O'Reilly Media, 2008.
- [7] D. C. Brown, "Decentering distortion of lenses," *Photogramm. Eng.*, vol. 32, no. 3, pp. 444–462, 1966.
- [8] A. E. Conrady, "Decentered lens systems," *Mon. Not. R. Astron. Soc.*, vol. 79, pp. 384–390, 1919.
- [9] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge: Cambridge Univ. Press, 2003.
- [10] J. Heikkila and O. Silven, "A four-step camera calibration procedure with implicit image correction," in *Proc. IEEE CVPR*, 1997, pp. 1106–1112.
- [11] K. Dutt and N. Kumar, "Digital Twin-Based Framework for Dynamic Stability Analysis of Grid-Tied Photovoltaic System," *IEEE Trans. Ind. Cyber-Physical Syst.*, 2025.
- [12] A. Kumar and N. Kumar, "Digital Twin Framework for 1-Phase Grid-Tied PV System," *IEEE Trans. Circuits Syst. I*, 2025.
- [13] R. Pandey and N. Kumar, "Digital Twin Formation and Real-Time Parameter Estimation," *IEEE Trans. Consum. Electron.*, 2025.
- [14] N. K. Kumawat and N. Kumar, "Comprehensive Component Health Monitoring," *IEEE Trans. Ind. Electron.*, 2025.
- [15] A. Kumar and N. Kumar, "State-Space Driven Digital Twin," *IEEE Trans. Ind. Cyber-Physical Syst.*, vol. 3, pp. 464–471, 2025.
- [16] R. Pandey and N. Kumar, "Enhanced Power Quality Control," *IEEE Trans. Ind. Appl.*, vol. 61, no. 1, pp. 676–685, 2025.
- [17] Raghav and N. Kumar, "Consumer-Focused Solutions," *IEEE Trans. Consum. Electron.*, vol. 71, no. 1, pp. 1784–1791, 2025.
- [18] N. Kumar, "EV Charging Adapter," *IEEE Trans. Energy Convers.*, vol. 39, no. 1, pp. 29–36, 2024.
- [19] N. Kumar and S. K. Panda, "Electric Vessels Charging Station," *IEEE Trans. Ind. Inform.*, vol. 19, no. 3, pp. 3254–3261, 2023.
- [20] N. Kumar and S. K. Panda, "Smart High Power Charging Networks," *IEEE Trans. Ind. Inform.*, vol. 19, no. 2, pp. 1476–1483, 2023.
- [21] S. Satapathy and N. Kumar, "Framework of MPPT for solar PV," *IET Renewable Power Gener.*, vol. 14, pp. 1668–1676, 2020.